Assignment 03: Condensed Matter Physics

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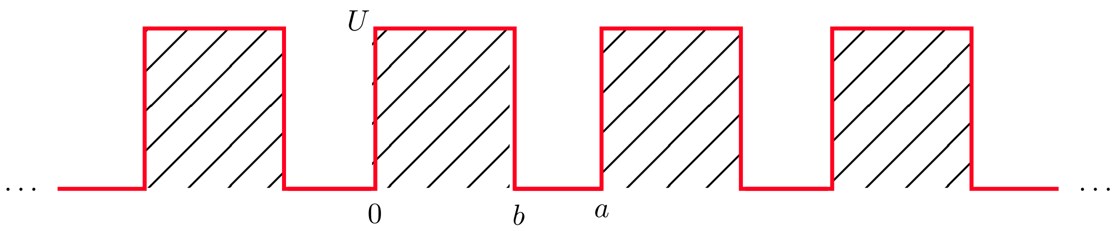
Solution 1: Kronig-Penney Model



(a) The dispersion relation obtained from the class for the Kronig-Penney model was

$$\cos(ka) = \cos(\alpha d)\cosh(\beta b) + \frac{\beta^2 - \alpha^2}{2\alpha\beta}\sin(\alpha d)\sinh(\beta b)$$

where $d = a - b$ and for simplicity we take $m = \hbar = 1$. We have the following

$$\triangleright\ \alpha = \sqrt{\frac{2mE}{\hbar^2}} = \sqrt{2E} \qquad \triangleright\ \beta = \sqrt{\frac{2m(U-E)}{\hbar^2}} = \sqrt{2(U-E)}$$

We numerically found the roots of the above dispersion relation, taking the parameters

$$\triangleright\ a = 1.0 \quad \triangleright\ b = 0.04 \quad \triangleright\ U = 80.0$$

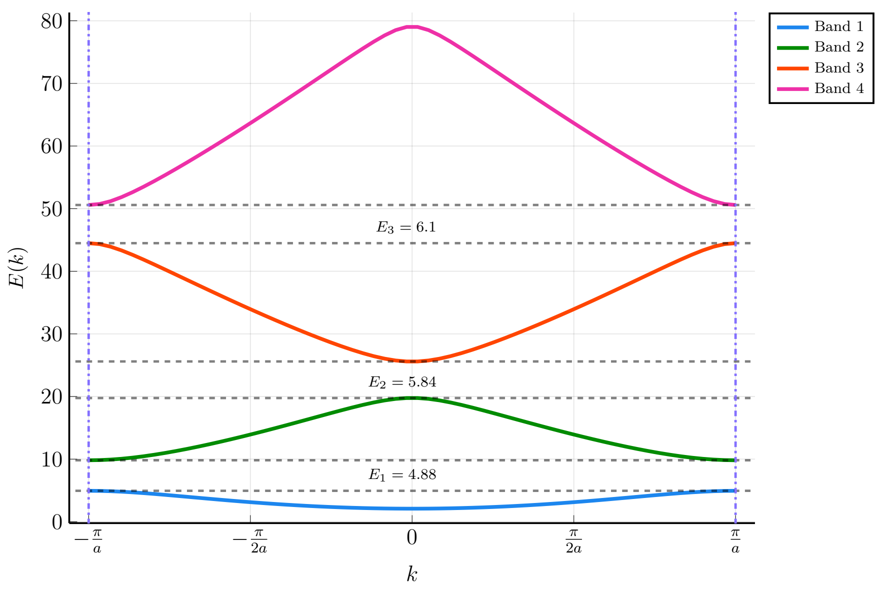


Figure 1: Energy dispersion for Kronig-Penney model. The figure shows the four lowest bands marked with different colors. The increasing bandgaps obtained for the above parameters are $\triangle E = 4.88, 5.84, 6.1$ denoted in the plot. The dashed horizontal lines denote the energy band boundaries and the vertical lines denote the BZ boundaries at $\pm^{\pi}/_{a}$

(b) Consider the dispersion relation for the KP model again

$$\cos(ka) = \cos(\alpha d)\cosh(\beta b) + \frac{\beta^2 - \alpha^2}{2\alpha\beta}\sin(\alpha d)\sinh(\beta b)$$

Taking $m = \hbar^2 = 1$, in the limit $U \to \infty$ and $b \to 0$ keeping $Ub = V_0$ constant, we have

$$\beta = \sqrt{2(U-E)} = \sqrt{2U}\left(1 - \frac{E}{2U}\right) \longrightarrow \beta b = \sqrt{2U}b - \frac{\sqrt{2}E}{\sqrt{U}} = \frac{\sqrt{2}(V_0 - E)}{\sqrt{U}} \xrightarrow{U\to\infty} 0$$

Hence the arguments of the hyperbolic trig functions can be taken to be 'small' and expanded upto linear order in βb from where we get: $\sinh(\beta b) \approx \beta b \qquad \cosh(\beta b) \approx 1$

Also note that $\sin(\alpha(a-b)) \approx \sin(\alpha a)$ and $\cos(\alpha(a-b)) \approx \cos(\alpha a)$ since $b \to 0$. Due to the same reason, we have

$$\frac{b}{2}(\beta^2 - \alpha^2) = \frac{b}{2}(2(U-E) - 2E) = b[U - 2E] = Ub - 2Eb = V_0 - 2Eb \xrightarrow{b\to 0} V_0$$

Substituting everything in the dispersion relation above we obtain:

$$\cos(ka) = \cos(\alpha d)\cosh(\beta b) + \frac{\beta^2 - \alpha^2}{2\alpha\beta}\sin(\alpha d)\sinh(\beta b) = \cos(\alpha a) \times 1 + \frac{\cancel{2}V_0}{\cancel{2}\cancel{b}\alpha\cancel{\beta}}\sin(\alpha a) \times (\cancel{\beta}\cancel{b})$$

Reintroducing terms for m and $\hbar$ we obtain

$$\boxed{\cos(ka) = \cos(\alpha a) + \frac{maV_0}{\hbar^2}\frac{\sin(\alpha a)}{\alpha a}}$$

We now take k to be complex and try to find the allowable energy solutions.

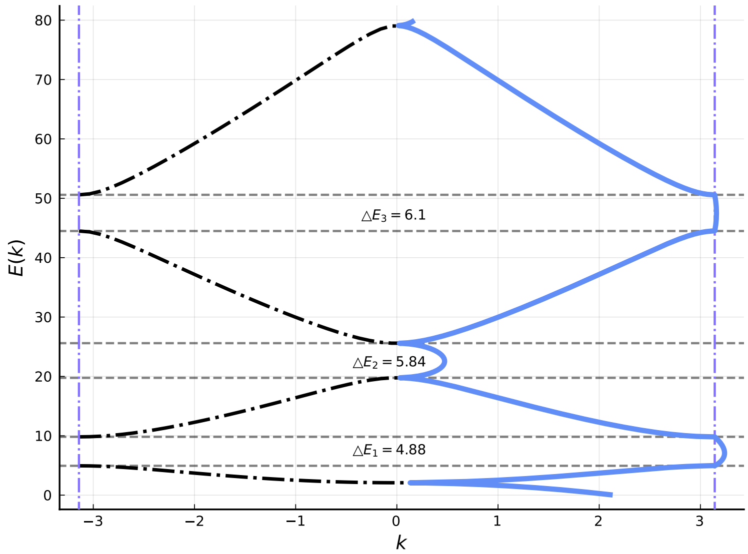


Figure 2: Dispersion for complex k. The dashed black lines corresponds to the bands previously obtain with real wave-vectors. The blue plot denote the energy solution vs. $|k|$ where $k \in \mathbb{C}$. We can see that for some $k \in \mathbb{C}$ we have E inside the forbidden region (bandgap).

Solution 2: 2D Square Lattice

(a) Let us consider the state with momentum ket $|\mathbf{k}\rangle \equiv |k_x, k_y\rangle$ and from Bloch's theorem, we can expand it as

$$|\mathbf{k}\rangle = \frac{1}{\sqrt{N}}\sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}}|\mathbf{R}\rangle \quad \longleftrightarrow \quad |k_x, k_y\rangle = \frac{1}{\sqrt{N}}\sum_{m,n} e^{ia(k_x m + k_y n)}|m,n\rangle$$

Now, substituting this in the Hamiltonian gives us

$$H|\mathbf{k}\rangle = -\frac{t}{\sqrt{N}}\sum_{m,n,p,q} e^{ia(k_x p + k_y q)}\Big\{ |m+1,n\rangle\langle m,n|p,q\rangle + |m,n+1\rangle\langle m,n|p,q\rangle + |m,n\rangle\langle m+1,n|p,q\rangle + |m,n\rangle\langle m,n+1|p,q\rangle \Big\} + \varepsilon_0 \sum_{m,n,p,q} e^{ia(k_x p + k_y q)}|m,n\rangle\langle m,n|p,q\rangle$$

Since the orbitals are localised we have the following condition: $\langle m,n|p,q\rangle = \delta_{mp}\delta_{nq} \quad \forall\ m,n,p,q$. Using this in the equation above we obtain:

$$
\begin{aligned}
H\left|\mathbf{k}\right\rangle &= -\frac{t}{\sqrt{N}}\sum_{p,q} e^{ia(k_{\mathrm{x}}p+k_{\mathrm{y}}q)}\Big\{\left|p+1,q\right\rangle + \left|p,q+1\right\rangle + \left|p-1,q\right\rangle + \\
&\qquad\qquad \left|p,q-1\right\rangle\Big\} + \varepsilon_0 \sum_{p,q} e^{ia(k_{\mathrm{x}}p+k_{\mathrm{y}}q)}\left|p,q\right\rangle \\
&= -\frac{t}{\sqrt{N}}\sum_{p,q}\Big(e^{ia(k_{\mathrm{x}}(p-1)+k_{\mathrm{y}}q)} + e^{ia(k_{\mathrm{x}}p+k_{\mathrm{y}}(q-1))} + e^{ia(k_{\mathrm{x}}(p+1)+k_{\mathrm{y}}q)} + e^{ia(k_{\mathrm{x}}p+k_{\mathrm{y}}(q+1))}\Big)\left|p,q\right\rangle + \varepsilon_0\left|\mathbf{k}\right\rangle \\
&= -t\Big(e^{-iak_{\mathrm{x}}} + e^{-iak_{\mathrm{y}}} + e^{iak_{\mathrm{x}}} + e^{iak_{\mathrm{y}}}\Big)\sum_{p,q}(e^{ia(k_{\mathrm{x}}p+k_{\mathrm{y}}q)})\left|p,q\right\rangle + \varepsilon_0\left|\mathbf{k}\right\rangle = [\varepsilon_0 - 2t\cos(k_{\mathrm{x}}a) - 2t\cos(k_{\mathrm{y}}a)]\left|\mathbf{k}\right\rangle
\end{aligned}
$$

We obtain an eigen-equation for $|\mathbf{k}\rangle$ and the dispersion relation thus obtained is

$$\boxed{E(\mathbf{k}) = [\varepsilon_0 - 2t\cos(k_{\mathrm{x}}a) - 2t\cos(k_{\mathrm{y}}a)]}$$

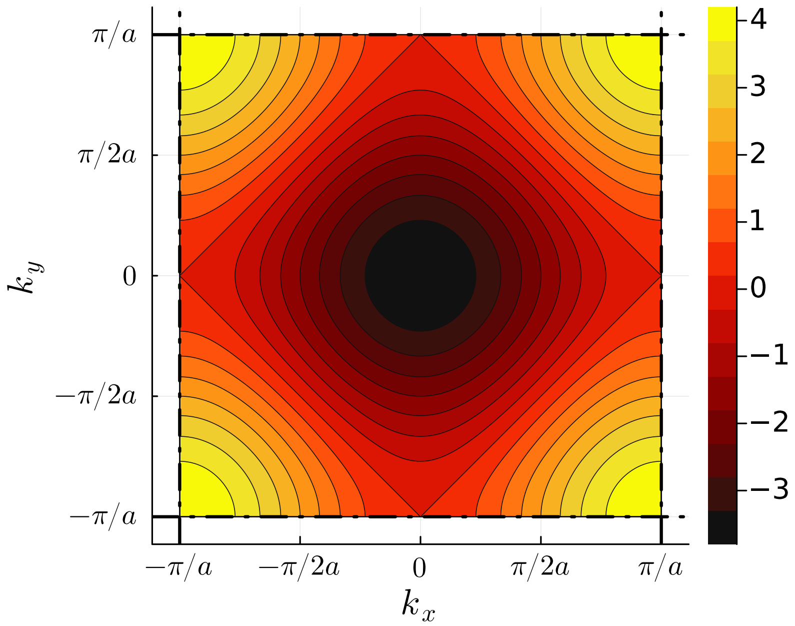


Figure 3: Dispersion for 2D Square lattice. Energy contours for 2D suare lattice within the first BZ; parameters used $t = 1.0, a = 1.0, \varepsilon_0 = 0.2$. The dashed lines denote the BZ boundary at $\pm\frac{\pi}{a}$

(b) The density of states is given by the expression

$$\rho(E) = \int\limits_{\mathbf{k}\in\mathrm{BZ}} \frac{\mathrm{d}^2k}{(2\pi)^2}\ \delta(E - E(\mathbf{k}))$$

Using the dispersion relation found out before, we have for the DOS

$$\rho(E) = \frac{1}{(2\pi)^2}\int\limits_{-\pi/a}^{\pi/a}\mathrm{d}k_{\mathrm{x}}\int\limits_{-\pi/a}^{\pi/a}\mathrm{d}k_{\mathrm{y}}\ \delta(E - [\varepsilon_0 - 2t\cos(k_{\mathrm{x}}a) - 2t\cos(k_{\mathrm{y}}a)])$$

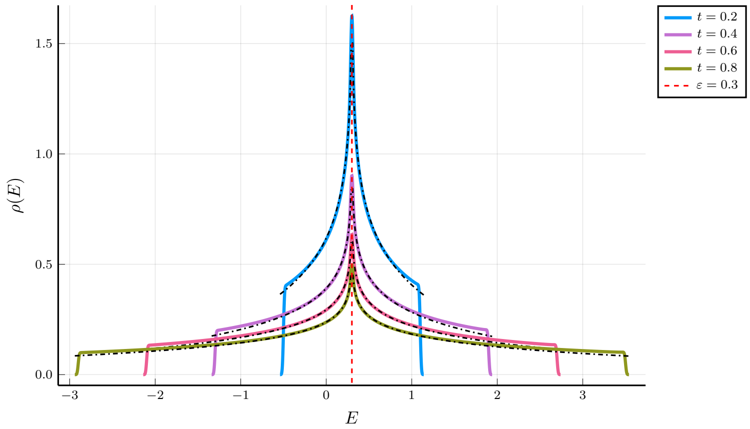


Figure 4: Density of states of 2D square lattice. The plots are shown for various values of the hopping amplitude and $\varepsilon_0 = 0.3$. We can clearly see a singularity of the DOS at $E = \varepsilon_0$. The black lines are fit for the DOS with the logarithmic divergence $A - B\ln|E - \varepsilon_0$

In 2D, the density of states can alternately be defined as,

$$g(\varepsilon) = \frac{1}{(2\pi)^2} \int_{\mathbf{k}' \in S(\varepsilon)} \frac{\mathrm{d}^2\mathbf{k}'}{|\nabla E(\mathbf{k}')|}$$

where $S(\varepsilon)$ is the constant energy surface, that is, $S(\varepsilon) = \{\mathbf{k}|E(\mathbf{k}) = \varepsilon\}$. From the expression above, we can see that the integrand shows a potential singularity at points where there is an extrememum of energy, that is, $\nabla E = 0$.

For our dispersion $E(k) = \varepsilon_0 - 2t(\cos(k_\mathrm{x}a) + \cos(k_\mathrm{y}a))$ we can see that:

$$\frac{\partial E}{\partial k_\mathrm{x}} = 2at\sin(k_\mathrm{x}a) = 0 \implies k_\mathrm{x} = 0, \pm\frac{\pi}{a}$$

Similar result holds for $\frac{\partial E}{\partial k_\mathrm{y}}$ too. However we leave the point $-\frac{\pi}{a}$ since this is equivalent to $\frac{\pi}{a}$ due to the periodicity of the BZ.

Now, we have potentially the points $\mathbf{k} = (0,0), (0,\pi/a), (\pi/a, 0), (\pi/a, \pi/a)$. Out of these, $(0,0)$ and $(\pi/a, \pi/a)$ are the minima and maxima since $E = \varepsilon_0 - 2t$ and $E = \varepsilon_0 + 2t$ respectively at these points, and singularities do not occur at these points. The other two points are the *saddle points* where the singularities occur and the DOS shows a logarithmic divergence around the singularity point, $E = \varepsilon_0 - 2t(-1+1) = \varepsilon_0$:

$$g(E) \sim -\ln|E - \varepsilon_0| \quad \text{where } E \text{ is close to } \varepsilon_0$$

(c) We consider the half-filling of electrons (with spin $\frac{1}{2}$) in this case. Thus for N sites we will have N momentum states with momentum $\mathbf{k}$ and we will have total $2N$ electrons occupying these states. Consider a transformation $\mathbf{k} \to \mathbf{k} + \left(\frac{\pi}{a}, \frac{\pi}{a}\right)$. The energy for this state is,

$$E(\mathbf{k}') = \varepsilon_0 - 2t\left[\cos\left(\left(k_\mathrm{x} + \frac{\pi}{a}\right)a\right) + \cos\left(\left(k_\mathrm{x} + \frac{\pi}{a}\right)a\right)\right] = \varepsilon_0 + 2t[\cos(k_\mathrm{x}a) + \cos(k_\mathrm{x}a)] = 2\varepsilon_0 - E(\mathbf{k})$$

We obtained a relation that for any such transformation $\mathbf{k} \to \mathbf{k} + \left(\frac{\pi}{a}, \frac{\pi}{a}\right)$, we have $E(\mathbf{k}') + E(\mathbf{k}) = 2\varepsilon_0$ Consider the density of states for the system,

$$g(E) = \frac{1}{(2\pi)^2} \int_{\mathbf{k} \in \mathrm{BZ}} \delta(E - E(\mathbf{k}))\ \mathrm{d}^2\mathbf{k}$$

Consider the following DOS in the periodic Brillouin zone:

$$g(2\varepsilon_0 - E) = \frac{1}{(2\pi)^2} \int_{\mathbf{k} \in \mathrm{BZ}} \delta(2\varepsilon_0 - E - E(\mathbf{k}))\ \mathrm{d}^2\mathbf{k}$$

$$\stackrel{\mathbf{k} \to \mathbf{k}'}{=} \frac{1}{(2\pi)^2} \int_{\mathbf{k}' \in \mathrm{BZ}} \delta(2\varepsilon_0 - E - E(\mathbf{k}'))\ \mathrm{d}^2\mathbf{k}' = \frac{1}{(2\pi)^2} \int_{\mathbf{k}' \in \mathrm{BZ}} \delta(E(\mathbf{k}) - E)\ \mathrm{d}^2\mathbf{k}'$$

We now use the fact that the delta function is symmetric about zero and the fact that $\int_{\mathbf{k}} \equiv \int_{\mathbf{k}'}$ from which we get that the DOS is symmetric about $2\varepsilon_0$:

$$g(2\varepsilon_0 - E) = \frac{1}{(2\pi)^2} \int_{\mathbf{k}\in \mathrm{BZ}} \delta(E - E(\mathbf{k}))\ \mathrm{d}^2\mathbf{k} = g(E)$$

Now consider the fully filled case for which we have:

$$2N = \int_{\varepsilon_0 - 4t}^{\varepsilon_0 + 4t} g(E)\mathrm{d}E \xrightarrow{E=\varepsilon_0 + x} \int_{-4t}^{4t} g(\varepsilon_0 + x)\ \mathrm{d}x$$

We break the above integral into two regions, above and below zero.

$$2N = \int_{-4t}^{0} g(\varepsilon_0 + x)\ \mathrm{d}x + \int_{4t}^{0} g(\varepsilon_0 + x)\ \mathrm{d}x$$

In the second integral we take $x \to -x$ and we get

$$2N = \int_{-4t}^{0} g(\varepsilon_0 + x)\ \mathrm{d}x + \int_{-4t}^{0} g(\varepsilon_0 - x)\ \mathrm{d}x$$

Finally we use the symmetry of the DOS around $2\varepsilon_0$ to get

$$2N = 2\int_{-4t}^{0} g(\varepsilon_0 + x)\ \mathrm{d}x \longrightarrow N = \int_{-4t}^{0} g(\varepsilon_0 + x)\ \mathrm{d}x \xrightarrow{y=\varepsilon_0 + x} \boxed{N = \int_{\varepsilon_0 - 4t}^{\varepsilon_0} g(y)\ \mathrm{d}y}$$

Note that for half-filling we have N electrons (out of $2N$ possible electrons occupying the states) and then we can fill this upto the Fermi-energy, say ε_{F} which has the expression,

$$\boxed{N = \int_{\varepsilon_0 - 4t}^{\varepsilon_{\mathrm{F}}} g(x)\ \mathrm{d}x}$$

Since the integrand (DOS) is positive, comparing the two boxed expressions we obtain $\varepsilon_{\mathrm{F}} = \varepsilon_0$. Now consider the Fermi momentum $\mathbf{k}_{\mathrm{F}} \equiv (K_{\mathrm{x}}, K_{\mathrm{y}})$ which satisfies,

$$\varepsilon_{\mathrm{F}} = \varepsilon_0 - 2t(\cos(K_{\mathrm{x}}a) + \cos(K_{\mathrm{y}}a)) \implies \boxed{\cos(K_{\mathrm{x}}a) + \cos(K_{\mathrm{y}}a) = 0}$$

This defines the Fermi surface for the 2D square lattice. We can simplify this further.

- $\cos(K_{\mathrm{x}}a) = -\cos(K_{\mathrm{y}}a) = \cos(K_{\mathrm{y}}a \pm \pi) \implies K_{\mathrm{x}}a = K_{\mathrm{y}}a \pm \pi + 2n\pi \implies (K_{\mathrm{x}} - K_{\mathrm{y}}) = \frac{\pi}{a}(2n \pm 1)$
- $\cos(K_{\mathrm{x}}a) = -\cos(-K_{\mathrm{y}}a) = \cos(-K_{\mathrm{y}}a \pm \pi) \implies K_{\mathrm{x}}a = -K_{\mathrm{y}}a \pm \pi + 2n\pi \implies (K_{\mathrm{x}} + K_{\mathrm{y}}) = \frac{\pi}{a}(2n \pm 1)$

Since the BZ $\equiv \left(-\frac{\pi}{a}.\frac{\pi}{a}\right) \times \left(-\frac{\pi}{a}.\frac{\pi}{a}\right)$ we can only have either :

$$K_{\mathrm{x}} \pm K_{\mathrm{y}} = \frac{\pi}{a} \quad \text{or.} \quad (K_{\mathrm{x}} \pm K_{\mathrm{y}}) = -\frac{\pi}{a}$$

This gives a Fermi surface which is a rhombus. Any state inside this rhombus will be filled and outside will not be filled.

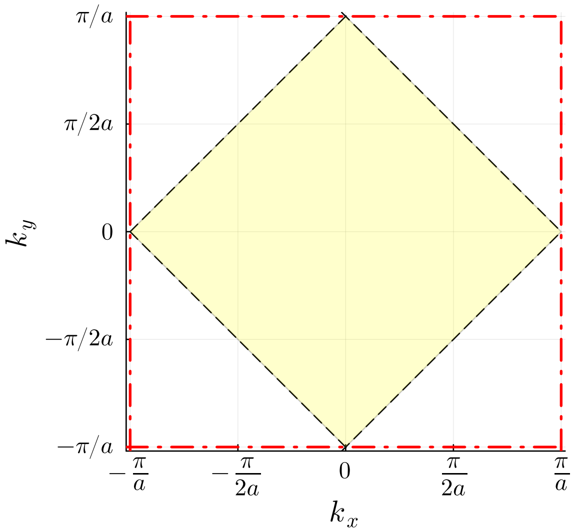
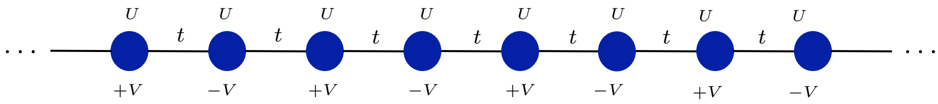


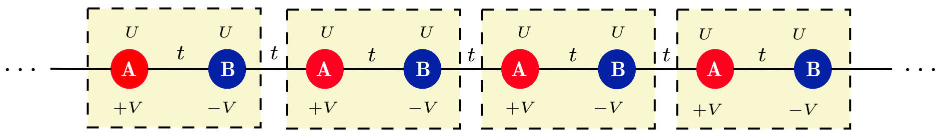
Figure 5: Fermi surface for square lattice. The red lines denote the BZ while the Fermi-surface is denoted by the black dashed lines. The yellow shaded region denotes the filled Fermi sea.

Solution 3: Tight Binding with Modulated On-site Potential

The given Hamiltonian can be schematically represented as below, where between each site, the hopping amplitude is $-t$ and the onsite-potential is U and an additional on-site potential of strength V is imposed with alternate signs along the sites.



We can equivalently divide the entire lattice into sublattices, where each unit cell now contains two atoms A and B, as shown below:



where the red sites in the j^{th} unit cell denote sublattice atoms of type A (located at sites $2j$) and the blue sites denote sublattice atoms of type B (located at sites $2j+1$). We can then define the lattice creation and annihilation operators for each sublattice sites viz. $\hat{c}^\dagger_{\mathrm{A,i}}$ and $\hat{c}_{\mathrm{A,i}}$ which will create and destroy particle at A site of unit cell i respectively and $\hat{c}^\dagger_{\mathrm{B,i}}$ and $\hat{c}_{\mathrm{B,i}}$ which will create and destroy particle at B site of unit cell i.

Dividing the lattice into two sublattices, we note that $+V$ $(-V)$ now occurs only at site A (site B) and then we can easily rewrite the above Hamiltonian as,

$$\hat{H} = -t\sum_i \left(\hat{c}^\dagger_{\mathrm{A,i+1}}\hat{c}_{\mathrm{B,i}} + \mathrm{h.c}\right) - t\sum_i \left(\hat{c}^\dagger_{\mathrm{A,i}}\hat{c}_{\mathrm{B,i}} + \mathrm{h.c}\right) + (U+V)\sum_i \hat{c}^\dagger_{\mathrm{A,i}}\hat{c}_{\mathrm{A,i}} + (U-V)\sum_i \hat{c}^\dagger_{\mathrm{A,i}}\hat{c}_{\mathrm{A,i}}$$

We can now expand these creation and annihilation operators in the momentum space as follows:

$$\hat{c}_{\mathrm{A,r}} = \frac{1}{\sqrt{N/2}}\sum_k e^{2ikra}\hat{c}_{\mathrm{A,k}} \qquad \hat{c}_{\mathrm{B,r}} = \frac{1}{\sqrt{N/2}}\sum_k e^{2ikra}\hat{c}_{\mathrm{B,k}}$$

We now substitute this in the Hamiltonian above. Let us do this term-by-term, first for the hopping terms:

$$\begin{aligned}
-t\sum_i \left(\hat{c}^\dagger_{\mathrm{A,i+1}}\hat{c}_{\mathrm{B,i}} + \hat{c}^\dagger_{\mathrm{B,i}}\hat{c}_{\mathrm{A,i+1}}\right) &= -\frac{t}{\sqrt{N/2}}\sum_{r,k,k'} e^{-2ika(r+1)}e^{2ik'ar}\hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{B,k'}} + e^{-2ik'ar}e^{2ika(r+1)}\hat{c}^\dagger_{\mathrm{B,k'}}\hat{c}_{\mathrm{A,k}} \\
&= -\frac{t}{\sqrt{N/2}}\sum_{r,k,k'} e^{-2ika}e^{-2ira(k-k')}\hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{B,k'}} + e^{2ika}e^{-2ira(k'-k)}\hat{c}^\dagger_{\mathrm{B,k'}}\hat{c}_{\mathrm{A,k}} \\
&= -t\sum_{k,k'} e^{-2ika}\delta_{k,k'}\hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{B,k'}} + e^{2ika}\delta_{k,k'}\hat{c}^\dagger_{\mathrm{B,k'}}\hat{c}_{\mathrm{A,k}} \\
&= -t\sum_{k} e^{-2ika}\hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{B,k}} + e^{2ika}\hat{c}^\dagger_{\mathrm{B,k}}\hat{c}_{\mathrm{A,k}}
\end{aligned}$$

$$\begin{aligned}
-t\sum_i \left(\hat{c}^\dagger_{\mathrm{A,i}}\hat{c}_{\mathrm{B,i}} + \hat{c}^\dagger_{\mathrm{B,i}}\hat{c}_{\mathrm{A,i}}\right) &= -\frac{t}{\sqrt{N/2}}\sum_{r,k,k'} e^{-2ikar}e^{2ik'ar}\hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{B,k'}} + e^{-2ik'ra}e^{2ikra}\hat{c}^\dagger_{\mathrm{B,k'}}\hat{c}_{\mathrm{A,k}} \\
&= -\frac{t}{\sqrt{N/2}}\sum_{r,k,k'} e^{-2ira(k-k')}\hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{B,k'}} + e^{-2ira(k'-k)}\hat{c}^\dagger_{\mathrm{B,k'}}\hat{c}_{\mathrm{A,k}} \\
&= -t\sum_{k,k'} \hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{B,k'}} + \hat{c}^\dagger_{\mathrm{B,k'}}\hat{c}_{\mathrm{A,k}} \\
&= -t\sum_{k} \hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{B,k}} + \hat{c}^\dagger_{\mathrm{B,k}}\hat{c}_{\mathrm{A,k}}
\end{aligned}$$

Similarly the on-site terms would have a similar form:

$$(U+V)\sum_i \hat{c}^\dagger_{\mathrm{A,i}}\hat{c}_{\mathrm{A,i}} = (U+V)\sum_k \hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{A,k}} \qquad (U-V)\sum_i \hat{c}^\dagger_{\mathrm{A,i}}\hat{c}_{\mathrm{A,i}} = (U-V)\sum_k \hat{c}^\dagger_{\mathrm{B,k}}\hat{c}_{\mathrm{B,k}}$$

where we have used the identity that $\frac{1}{\sqrt{N/2}}\sum_r e^{-2ia(k-k')r} = \delta_{k,k'}$. Now joining these terms together we get the following:

$$\hat{H} = \sum_k -t(1+e^{-2ika})\hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{B,k}} - t(1+e^{2ika})\hat{c}^\dagger_{\mathrm{B,k}}\hat{c}_{\mathrm{A,k}}(U+V)\hat{c}^\dagger_{\mathrm{A,k}}\hat{c}_{\mathrm{A,k}} + (U-V)\hat{c}^\dagger_{\mathrm{B,k}}\hat{c}_{\mathrm{B,k}}$$

Let us define the vector $\hat{\mathbf{c}}_{\mathrm{k}} := \begin{pmatrix}\hat{c}_{\mathrm{A,k}} \\ \hat{c}_{\mathrm{B,k}}\end{pmatrix}$ and then we can compactly write the Hamiltonian as:

$$\hat{H} = \sum_k \hat{\mathbf{c}}^\dagger_{\mathrm{k}}\ h(k)\ \hat{\mathbf{c}}_{\mathrm{k}}$$

where the matrix $h(k)$ takes the following form:

$$h(k) = \begin{pmatrix} (U+V) & -t(1+e^{-2ika}) \\ -t(1+e^{2ika}) & (U-V)\end{pmatrix} \xrightarrow{\text{diagonalise}} \det\begin{pmatrix} (U+V-\lambda) & -t(1+e^{-2ika}) \\ -t(1+e^{2ika}) & (U-V-\lambda)\end{pmatrix} = 0$$

We obtain the characteristic equation as:

$$(U-\lambda)^2 - V^2 - t^2(2+2\cos(2ka)) = 0 \implies (U-\lambda)^2 = V^2 + 4t^2\cos^2(ka)$$

The dispersion relation obtained for the modulated on-site potential is

$$\boxed{\lambda_\pm(k) = U \pm \sqrt{V^2 + 4t^2\cos^2(ka)}}$$

Since we have two sublattices, we now get a two-band dispersion relation. (b) If $V = 0$ then the lattice has translational symmetry of unit a, however, when $V \neq 0$ then this translational symmetry becomes $2a$ (which we also saw explicitly from the sublattice structure). Thus the Brillouin zone reduces from $(-\pi/a, \pi/a)$ to $(-\pi/2a, \pi/2a)$ due to the introduction of the staggered potential.

Solution 4: Ladder Lattice

The lattice can be schematically shown as:

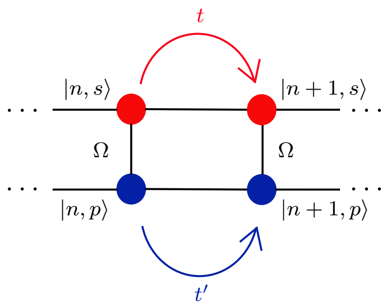


Figure 6: Schematic representation of two coupled 1D chains corresponding to s and p orbitals. The hopping amplitudes along the s and p chains are t and t' respectively, while the inter-chain coupling (hybridization) is Ω.

We can expand the orbital states in terms of the Fourier modes as,

$$|n,\sigma\rangle = \frac{1}{\sqrt{N}}\sum_k e^{-ikna}|k,\sigma\rangle$$

where σ denotes the orbital type ($\sigma\in\{$s,p$\}$). Let us consider the terms of the Hamiltonian one by one.

$$\begin{aligned}
-t\sum_n |n,s\rangle\langle n+1,s| + |n+1,s\rangle\langle n,s| &= -\frac{t}{N}\sum_{n,k,k'}(e^{-ikna}e^{ik'(n+1)a} + e^{-ik(n+1)a}e^{ik'na})\,|k,s\rangle\langle k',s| \\
&= -\frac{t}{N}\sum_{n,k,k'}(e^{ik'a}e^{-ina(k-k')} + e^{-ika}e^{-ina(k-k')})\,|k,s\rangle\langle k',s| \\
&= -t\sum_{k,k'}(e^{ik'a}\delta_{k,k'} + e^{-ika}\delta_{k,k'})\,|k,s\rangle\langle k',s| \\
&= -t\sum_k(e^{ika}+e^{-ika})\,|k,s\rangle\langle k,s| = \sum_k(-2t\cos(ka))\,|k,s\rangle\langle k,s|
\end{aligned}$$

Similarly, the second term in the Hamiltonian leads to

$$-t'\sum_n |n,p\rangle\langle n+1,p| + |n+1,p\rangle\langle n,p| = \sum_k(-2t'\cos(ka))\,|k,p\rangle\langle k,p|$$

Let us now consider the onsite terms in the Hamiltonian:

$$\varepsilon_{\mathrm{s}}\sum_n |n,s\rangle\langle n,s| = \frac{\varepsilon_{\mathrm{s}}}{N}\sum_{n,k,k'} e^{-ikna}e^{ik'na}\,|k,s\rangle\langle k',s| = \frac{\varepsilon_{\mathrm{s}}}{N}\sum_{n,k,k'} e^{-ina(k-k')}\,|k,s\rangle\langle k',s| = \varepsilon_{\mathrm{s}}\sum_k |k,s\rangle\langle k,s|$$

Similarly, for the p orbital we will have the onsite term as:

$$\varepsilon_{\mathrm{p}}\sum_n |n,p\rangle\langle n,p| = \varepsilon_{\mathrm{p}}\sum_k |k,p\rangle\langle k,p|$$

Finally, let us consider the hopping term between the parallel chains:

$$\begin{aligned}
\Omega\sum_n |n,s\rangle\langle n,p| + |n,p\rangle\langle n,s| &= -\frac{\Omega}{N}\sum_{n,k,k'} e^{-ikna}e^{ik'na}\,|k,s\rangle\langle k',p| + e^{-ikna}e^{ik'na}\,|k,p\rangle\langle k',s| \\
&= -\frac{\Omega}{N}\sum_{n,k,k'} e^{-i(k-k')na}\big[|k,s\rangle\langle k',p| + |k,p\rangle\langle k',s|\big] \\
&= -\Omega\sum_k |k,s\rangle\langle k,p| + |k,p\rangle\langle k,s|
\end{aligned}$$

Now joining all these terms together we get the Hamiltonian expanded in the momentum space as,

$$\hat{H} = \sum_k (\varepsilon_p - 2t' \cos(ka)) |k,p\rangle \langle k,p| + (\varepsilon_s - 2t\cos(ka)) |k,s\rangle \langle k,s| - \Omega |k,s\rangle \langle k,p| - \Omega |k,p\rangle \langle k,s|$$

Now using the fact that $\langle q,\sigma|q',\sigma'\rangle = \delta_{\sigma\sigma'}\delta_{q,q'}$ we can obtain the matrix elements:

- $\triangleright$ $\langle k',s|\hat{H}|k',s\rangle = \varepsilon_s - 2t\cos\big(k'a\big)$ $\triangleright$ $\langle k',s|\hat{H}|k',p\rangle = -\Omega$
- $\triangleright$ $\langle k',p|\hat{H}|k',p\rangle = \varepsilon_p - 2t'\cos\big(k'a\big)$ $\triangleright$ $\langle k',p|\hat{H}|k',s\rangle = -\Omega$

$$H(k) = \begin{pmatrix} \varepsilon_s - 2t\cos(ka) & -\Omega \\ -\Omega & \varepsilon_p - 2t'\cos(ka) \end{pmatrix}$$

(b) The energy dispersion is found from the eigenvalues of the above matrix $H(k)$. We obtain the characteristic equation using

$$\det \begin{pmatrix} \varepsilon_s - 2t\cos(ka) - \lambda & -\Omega \\ -\Omega & \varepsilon_p - 2t'\cos(ka) - \lambda \end{pmatrix} = 0$$

Let us denote $h_1 = 2t\cos(ka)$ and $h_2 = 2t'\cos(ka)$ for simplicity. Then we have the equation as,

$$(\lambda - \varepsilon_s + h_1)(\lambda - \varepsilon_p + h_2) - \Omega^2 = 0 \implies \lambda^2 - \lambda((\varepsilon_p + \varepsilon_s) - (h_1 + h_2)) + (h_1 - \varepsilon_s)(h_2 - \varepsilon_p) - \Omega^2 = 0$$

The solutions to this quadratic equation are given as:

$$\boxed{E_\pm = \frac{1}{2}\Big\{((\varepsilon_p + \varepsilon_s) - (h_1 + h_2)) \pm \sqrt{((\varepsilon_p + \varepsilon_s) - (h_1 + h_2))^2 - 4[(h_1 - \varepsilon_s)(h_2 - \varepsilon_p) - \Omega^2]}\Big\}}$$

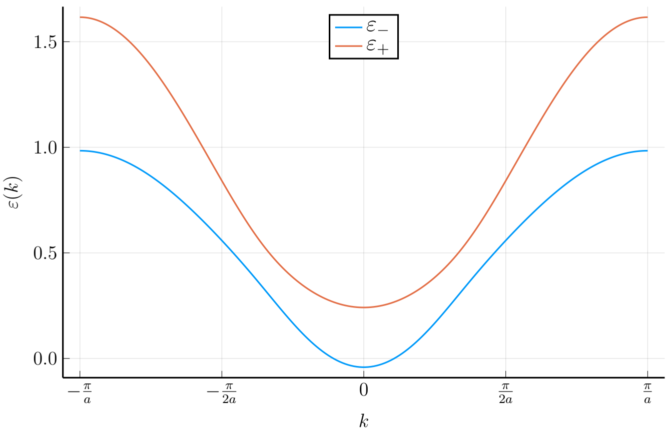


Figure 7: Dispersion for the ladder lattice. The parameters used are $a = 1, \varepsilon_s = 0.8, \varepsilon_p = 0.6, t = 0.4, t' = 0.2, \Omega = 0.1$

Solution 5: Bloch's Theorem

(a) Before substituting the equation in time-independent Schrödinger, note that

$$\hat{p} \equiv -i\hbar\partial_x \implies \hat{p}e^{ikx} = (-i\hbar)(ik)e^{ikx} = (\hbar k)e^{ikx}$$

Using this fact we can easily find that,

$$\begin{aligned} \hat{p}^2(e^{ikx}u_{n,k}(x)) &= \hat{p}\Big((\hbar k)e^{ikx}u_{n,k} + e^{ikx}\hat{p}u_{n,k}\Big) \\ &= (\hbar k)\Big[(\hbar k)e^{ikx}u_{n,k} + e^{ikx}\hat{p}u_{n,k}\Big] + (\hbar k)e^{ikx}\hat{p}u_{n,k} + e^{ikx}\hat{p}^2u_{n,k} \\ &= e^{ikx}[(\hbar k)^2u_{n,k} + 2(\hbar k)\hat{p}u_{n,k} + \hat{p}^2u_{n,k}] \equiv e^{ikx}(\hbar k + \hat{p})^2u_{n,k} \end{aligned}$$

Hence we have

$$\hat{H}\psi_{n,k}(x) = \frac{\hat{p}^2}{2m}\psi_{n,k} + V(x)\psi_{n,k} = \cancel{e^{ikx}}\underbrace{\left[\frac{1}{2m}(\hbar k + \hat{p})^2 + V(x)\right]}_{h_k}u_{n,k} = \varepsilon_{\mathrm{n,k}}\cancel{e^{ikx}}u_{n,k} \implies \boxed{h_k u_{\mathrm{n,k}}(x) = \varepsilon_{\mathrm{n,k}}u_{\mathrm{n,k}}(x)}$$

(b) We now consider a perturbation with a weak potential such that only the lowest harmonic contributes to $V(x)$. We then have,

$$V(x) = V_{\mathrm{G}_1}e^{iG_1x} + V^*_{\mathrm{G}_1}e^{-iG_1x}$$

where we have, for a general potential, $V(x) = \sum\limits_{\mathrm{G}} V_{\mathrm{G}}e^{i\mathrm{G}x}$ and $G = {}^{2\pi n}/_{a}$ are the reciprocal lattice vectors. In the absence of the potential we have plane-wave solutions and the dispersion is quadratic, $\varepsilon(k) \sim k^2$ which implies that at the Brillouin zone boundaries $k = \pm{}^{\pi}/_{a} = \pm{}^{G_1}/_{2}$, we will have a degeneracy and thus we have to apply degenerate perturbation theory at the BZ boundaries! The unperturbed wavefunctions and energy at $\pm{}^{G_1}/_{2}$ are:

$$\phi_1(x) = \frac{1}{\sqrt{L}}e^{i(G_1/2)x} \qquad \phi_2(x) = \frac{1}{\sqrt{L}}e^{-i(G_1/2)x} \qquad ; \quad \varepsilon_0 = \frac{\hbar^2 G_1^2}{8m}$$

where $L = Na$ is the length of the lattice. We have to diagonalise the potential in the degenerate subspace. For that consider $k = {}^{G_1}/_{2}$ and $k' = {}^{-G_1}/_{2}$

$$\langle k|\, H\, |k\rangle = \frac{1}{L}\int\limits_0^L e^{-i(G_1/2)x}(\varepsilon_0 + V_{\mathrm{G}_1}e^{iG_1x} + V^*_{\mathrm{G}_1}e^{-iG_1x})e^{i(G_1/2)x}\, \mathrm{d}x$$

$$= \varepsilon_0 + \frac{V_{\mathrm{G}_1}}{L}\left[\frac{e^{iG_1x}}{iG_1}\right]_0^L + \frac{V^*_{\mathrm{G}_1}}{L}\left[\frac{e^{-iG_1x}}{-iG_1}\right]_0^L = \varepsilon_0$$

since $G_1L = \frac{2\pi}{a} \times Na = 2\pi N \in \mathbb{N}$ and $\exp(2\pi ni) = 1\ \forall\ n \in \mathbb{Z}$. Similarly we have $\langle k'|\, V\, |k'\rangle = \varepsilon_0$ by a similar calculation. Let us now consider the off-diagonal terms.

$$\langle k'|\, H\, |k\rangle = \frac{1}{L}\int\limits_0^L e^{i(G_1/2)x}(\varepsilon_0 + V_{\mathrm{G}_1}e^{iG_1x} + V^*_{\mathrm{G}_1}e^{-iG_1x})e^{i(G_1/2)x}\, \mathrm{d}x$$

$$= \frac{1}{L}\int\limits_0^L e^{iG_1x}(\varepsilon_0 + V_{\mathrm{G}_1}e^{iG_1x} + V^*_{\mathrm{G}_1}e^{-iG_1x})\, \mathrm{d}x = V^*_{\mathrm{G}_1}$$

where the first two terms are zero (as shown before in the previous calculation) and the last term is only non-zero. We thus get $\langle k'|\, H\, |k\rangle = V^*_{\mathrm{G}_1}$. Taking the Hermitian conjugate of this equation we have $\langle k|\, H\, |k'\rangle = V_{\mathrm{G}_1}$. Then we get the effective matrix in the degenerate subspace as:

$$H_{\mathrm{sub}} = \begin{pmatrix} \varepsilon_0 & V_{\mathrm{G}_1} \\ V^*_{\mathrm{G}_1} & \varepsilon_0 \end{pmatrix}$$

The eigenvalues of the matrix are found by solving:

$$(\lambda - \varepsilon_0)^2 = |V_{\mathrm{G}_1}|^2 \implies \lambda = \varepsilon_0 \pm |V_{\mathrm{G}_1}|$$

Thus we see that a gap opens up at the zone boundary and $\Delta = 2|V_{\mathrm{G}_1}|$ is the gap between the energies obtained after the perturbation.

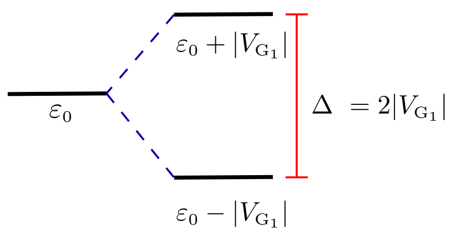


Figure 8: Splitting of energy levels on introducing the perturbative weak-potential.

(c) We have been given a periodic Gaussian potential

$$V(x) = -V_0 \sum_n e^{-\alpha(x-na)^2}$$

We have to find the Fourier coefficients V_{G} of this potential which are obtained as:

$$V_{\mathrm{G}} = \frac{1}{a}\int_0^a \mathrm{d}x \left[-V_0 \sum_{n=-\infty}^{\infty} e^{-\alpha(x-na)^2}\right] e^{-iGx} = -\frac{V_0}{a}\int_0^a \mathrm{d}x \left[\sum_{n=-\infty}^{\infty} e^{-\alpha(x-na)^2}\right] e^{-iGx}$$

Let us substitute $y = x - na \longrightarrow \mathrm{d}y = \mathrm{d}x$. Then the coefficients becomes

$$V_{\mathrm{G}} = -\frac{V_0}{a}\sum_{n=-\infty}^{\infty}\int_{-na}^{a(1-n)} \mathrm{d}y\; e^{-\alpha y^2} e^{-iG(y+na)}$$

Note that the integral is over a region of width a and for different values of n, we obtain non-overlapping regions of width a. Thus, essentially, the sum and integral together forms the entire real line $\mathbb{R}$ and we can just replace the sum+integral with just an integral over the real line.

Also note that the reciprocal lattice vectors are of the form $G = {}^{2\pi m}/_{a}$ and hence $Gna = 2\pi mn \implies e^{-iGna} = 1$. Using these facts, we can write the coefficients as

$$V_{\mathrm{G}} = -\frac{V_0}{a}\int_{-\infty}^{\infty} \mathrm{d}y\; e^{-\alpha y^2} e^{-iGy}$$

The integral evaluates to $\sqrt{\frac{\pi}{\alpha}} e^{-G^2/4\alpha}$ and thus we get

$$\boxed{V_{\mathrm{G}} = -\frac{V_0}{a}\sqrt{\frac{\pi}{\alpha}} e^{-G^2/4\alpha}}$$

Then from the expression found out previously, for zone-boundary ($G = {}^{2\pi}/_{a}$) the band-gap becomes,

$$\boxed{\Delta = 2|V_{\mathrm{G}_1}| = \frac{2V_0}{a}\sqrt{\frac{\pi}{\alpha}}\; e^{-\pi^2/a^2\alpha}}$$

(d) Let us consider a complex wavevector of the form $k = \frac{G_1}{2} + i\chi$, which gives us

$$\varepsilon_0(\pm k) = \frac{\hbar^2}{2m}\left[\pm\frac{G_1}{2} + i\chi\right]^2 = \frac{\hbar^2}{2m}\left[\left(\frac{G_1}{2}\right)^2 - \chi^2 \pm i\chi G_1\right]$$

The energies at the BZ boundaries now have a *complex part*, however considering $\chi \ll G_1$ we can still use the perturbation theory. The Hamiltonian in the denegerate subspace now changes as:

$$H_{\mathrm{sub}} = \begin{pmatrix} \varepsilon_0^+ & V_{\mathrm{G}_1} \\ V_{\mathrm{G}_1}^* & \varepsilon_0^- \end{pmatrix}$$

The eigenvalues are now found out from the characteristic equation,

$$\begin{aligned}
\implies\quad & (\lambda - \varepsilon_0^+)(\lambda - \varepsilon_0^-) - |V_{\mathrm{G}_1}|^2 = 0 \\
\implies\quad & \lambda^2 - \lambda(\varepsilon_0^+ + \varepsilon_0^-) + \varepsilon_0^+\varepsilon_0^- - |V_{\mathrm{G}_1}|^2 = 0 \\
\implies\quad & \lambda_\pm = \frac{1}{2}\left\{(\varepsilon_0^+ + \varepsilon_0^-) \pm \sqrt{(\varepsilon_0^+ + \varepsilon_0^-)^2 - 4(\varepsilon_0^+\varepsilon_0^- - |V_{\mathrm{G}_1}|^2)}\right\} \\
& = \frac{1}{2}\left\{(\varepsilon_0^+ + \varepsilon_0^-) \pm \sqrt{(\varepsilon_0^+ - \varepsilon_0^-)^2 + 4|V_{\mathrm{G}_1}|^2}\right\} \\
& = \frac{\hbar^2}{2m}\left[\left(\frac{G_1}{2}\right)^2 - \chi^2\right] \pm \sqrt{|V_{\mathrm{G}_1}|^2 - \left(\frac{\hbar^2}{2m}\chi G_1\right)^2}
\end{aligned}$$

By physical grounds, $\lambda \in \mathbb{R}$ and thus we have the following

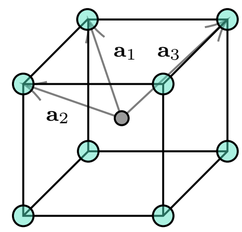
$$|V_{G_1}|^2 \geq \left(\frac{\hbar^2}{2m}\chi G_1\right)^2 \implies \chi \leq \frac{2m|V_{G_1}|}{\hbar^2 G_1} \quad \text{or.} \quad \chi \geq -\frac{2m|V_{G_1}|}{\hbar^2 G_1}$$

Suppose we choose $\chi = \frac{2m|V_{G_1}|}{\hbar^2 G_1}$ from which we get $\lambda_\pm = \frac{\hbar^2}{2m}\left(\frac{G_1}{2}\right)^2 - \frac{2m}{\hbar^2}\left(\frac{|V_{G_1}|}{G_1}\right)^2$, which for 'small' enough $|V_{G_1}|/G_1$, is equal to the energy without splitting in part (b). For any χ, note that the square root term is always less than $|V_{G_1}|$ and hence the splitting will be lesser than what was in part (b) and this state will be in the *forbidden region.* Thus for any $k \in \mathbb{C}$ defined with appropriate chosen value of ξ, there is a continuous set of allowed energies inside the gap, parameterized by χ.

Solution 6: Reciprocal Lattice

► Body-centered cubic

Let us consider the primitive unit cell for bcc lattice, such that each side of the cube is a and let us consider the atom at the centre of the cube to be the origin.



There are 8 nearest neighbours of this atom and we choose three of them to be the *primitive lattice vectors.*

$$\mathbf{a}_1 = \frac{a}{2}\begin{pmatrix}-1\\-1\\1\end{pmatrix} \qquad \mathbf{a}_2 = \frac{a}{2}\begin{pmatrix}1\\-1\\1\end{pmatrix} \qquad \mathbf{a}_3 = \frac{a}{2}\begin{pmatrix}-1\\1\\1\end{pmatrix}$$

We calculate the volume $v = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3) = a^3/2$. From this we can calculate the *reciprocal lattice vectors*,

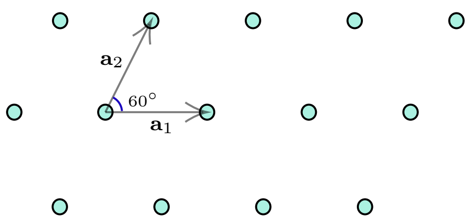
$$\mathbf{b}_1 = \frac{2\pi(\mathbf{a}_2 \times \mathbf{a}_3)}{v} = \frac{2\pi}{a}\begin{pmatrix}-1\\-1\\0\end{pmatrix}$$

$$\mathbf{b}_2 = \frac{2\pi(\mathbf{a}_3 \times \mathbf{a}_1)}{v} = \frac{2\pi}{a}\begin{pmatrix}1\\0\\1\end{pmatrix}$$

$$\mathbf{b}_3 = \frac{2\pi(\mathbf{a}_1 \times \mathbf{a}_2)}{v} = \frac{2\pi}{a}\begin{pmatrix}0\\1\\1\end{pmatrix}$$

► Triangular Lattice

Let us consider the primitive unit cell for the 2D triangular lattice, such that each side of the triangle is a.



There are 6 nearest neighbours of an atom and we choose two of them to be the *primitive lattice vectors.*

$$\mathbf{a}_1 = a\begin{pmatrix}1\\0\end{pmatrix} \qquad \mathbf{a}_2 = \frac{a}{2}\begin{pmatrix}1\\\sqrt{3}\end{pmatrix}$$

From the orthogonality relation we can say that the reciprocal lattice vectors, $\mathbf{b}_1 \perp \mathbf{a}_2$ and $\mathbf{b}_2 \perp \mathbf{a}_1$. Then suppose $\mathbf{b}_1 = c(\sqrt{3}, -1)^{\mathrm{T}}$ and $\mathbf{b}_2 = d(0, -1)^{\mathrm{T}}$. Moreover, from the normalisation we have,

$$\mathbf{a}_1 \cdot \mathbf{b}_1 = 2\pi \implies c = \frac{2\pi}{\sqrt{3}a}$$
$$\mathbf{a}_2 \cdot \mathbf{b}_2 = 2\pi \implies d = -\frac{4\pi}{\sqrt{3}a}$$

Thus we get the *reciprocal lattice vectors* as

$$\mathbf{b}_1 = \frac{2\pi}{\sqrt{3}a}\begin{pmatrix}\sqrt{3}\\ -1\end{pmatrix} \qquad \mathbf{b}_2 = \frac{4\pi}{\sqrt{3}a}\begin{pmatrix}0\\ 1\end{pmatrix}$$

Solution 7: Ultracold Atoms

Note that the minima of the given potential occurs at $x_n = \left(n + \frac{1}{2}\right)a$ and we can expand the potential about this minima, where from the Taylor series around x_n, we get,

$$V(x) = V(x_n) + \left.\frac{\mathrm{d}V}{\mathrm{d}x}\right|_{x_n}(x - x_n) + \frac{1}{2}\left.\frac{\mathrm{d}^2V}{\mathrm{d}x^2}\right|_{x_n}(x - x_n)^2 + \ldots$$

The first term vanishes as the minimum value of the potential is zero. The second term vanishes since at minima, the slope is zero. The quadratic term is:

$$\frac{1}{2}\left.\frac{\mathrm{d}^2V}{\mathrm{d}x^2}\right|_{x_n} = \frac{V_0}{2}\frac{\mathrm{d}}{\mathrm{d}x}\left.\left(-\frac{2\pi}{a}\cos\left(\frac{\pi x}{a}\right)\sin\left(\frac{\pi x}{a}\right)\right)\right|_{x_n} = -\frac{\pi V_0}{2a}\frac{\mathrm{d}}{\mathrm{d}x}\left.\left(\sin\left(\frac{2\pi x}{a}\right)\right)\right|_{x_n} = -\frac{\pi^2 V_0}{a^2}\cos\left.\left(\frac{2\pi x}{a}\right)\right|_{x_n} = \frac{\pi^2}{a^2}V_0$$

Then we have an effective harmonic oscillator potential around each minima:

$$V(x) = \frac{\pi^2}{a^2}V_0(x - x_n)^2$$

Let us map the parameters with the usual SHO parameters:

$$\frac{1}{2}m\omega^2 = \frac{\pi^2}{a^2}V_0 \implies \omega = \frac{\pi}{a}\sqrt{\frac{2V_0}{m}}$$

To construct the localised wavefunctions to expand the Bloch state, we can choose the ground state of the harmonic oscillator, since this potential is effectively harmonic.

$$\varphi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega(x-x_n)^2/2\hbar} \longrightarrow \boxed{\varphi_n(x) = \frac{1}{(\pi\sigma^2)^{1/4}}e^{-(x-x_n)^2/2\sigma^2}}$$

where $\sigma = \sqrt{\hbar/m\omega}$ is the characteristic length scale of the system. We can construct the Bloch wavefunction $\Psi_k(x)$ using a linear combination of these localised *Wannier* functions in the tight-binding approximation.

$$\Psi_k(x) = \frac{1}{\sqrt{N}}\sum_n e^{ikna}\varphi_n(x) \quad ; \quad \varphi_n(x) = \frac{1}{(\pi\sigma^2)^{1/4}}\exp\left(-\frac{(x - x_n)^2}{2\sigma^2}\right)$$

(b) Let us now calculate the tunneling amplitude t which represents the probability of hopping between sites. Within the tight-binding approximation we only need to consider the nearest neighbour hopping and we have

$$t \approx -\langle\varphi_n|\hat{H}|\varphi_{n+1}\rangle = -\int_{-\infty}^{\infty}\varphi_n^*(x)\left(-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)\right)\varphi_{n+1}(x)$$

Now consider the Hamiltonian for a harmonic oscillator centred at x_{n+1} which can be written as:

$$\hat{H}_{n+1} = \hat{T} + \frac{1}{2}m\omega^2(x - x_{n+1})2 \implies \hat{H} = \hat{H}_{n+1} + (V - V_{n+1}) \equiv \hat{H}_{n+1} + \delta V_{n+1}$$

We can substitute this in the above integral and we obtain:

$$t \approx -\langle\varphi_n|\hat{H}_{n+1}|\varphi_{n+1}\rangle - \langle\varphi_n|\delta V_{n+1}|\varphi_{n+1}\rangle$$

Note that φ_{n+1} is the ground state of $\hat{H}_{n+1}$ and thus the first term yields $-\varepsilon_0\langle\varphi_n|\varphi_{n+1}\rangle$ which is just the overlap between two Gaussian functions. Now let us come to the second term:

$$-\int_{-\infty}^{\infty} \varphi_n^*(x)\ \delta V_{n+1}(x)\ \varphi_{n+1}(x)$$

Note that, in the overlap region, that is, for x between x_n and x_{n+1}, the wavefunctions are negligible (since these are localised at the centres), although δV is large (since there is significant deviation from the harmonic approximation), hence the integrand is negligible.

Moreover, around the centre x_{n+1}, the wavefunction φ_{n+1} is large but δV_{n+1} is small, since the harmonic approximation holds good around the centre. Around the centre x_n, δV_{n+1} is large, but the wavefunction φ_{n+1} is small. Hence in the region of importance, the integral is negligible and hence we can ignore this term from the tunneling amplitude.

$$\langle\varphi_n|\varphi_{n+1}\rangle = \frac{1}{\sqrt{\pi\sigma^2}}\int_{-\infty}^{\infty} \exp\left[-\frac{(x-x_n)^2}{2\sigma^2}\right]\exp\left[-\frac{(x-x_{n+1})^2}{2\sigma^2}\right]$$

Note the following:

$$\begin{aligned}(x-A)^2 + (x-B)^2 = 2(x^2 - x(A+B) + \frac{1}{2}(A^2+B^2)) &= 2\left\{\left[x - \left(\frac{A+B}{2}\right)\right]^2 - \left(\frac{A+B}{2}\right)^2 + \frac{A^2+B^2}{2}\right\} \\ &= 2\left\{\left[x - \left(\frac{A+B}{2}\right)\right]^2 - \frac{(A-B)^2}{4}\right\}\end{aligned}$$

In our case we have $A = \left(n + \frac{1}{2}\right)a$ and $B = \left(n + \frac{3}{2}\right)a$ from which we get $A+B = 2(n+1)a$ and $B - A = a$. Then the overlap integral becomes:

$$\frac{1}{\sqrt{\pi\sigma^2}}\int_{-\infty}^{\infty} \exp\left[-\frac{1}{\sigma^2}\left\{(x-(n+1)a)^2 - \frac{a^2}{4}\right\}\right] = \frac{e^{-a^2/4\sigma^2}}{\sqrt{\pi\sigma^2}}\int_{-\infty}^{\infty} \exp\left[-\frac{1}{\sigma^2}(x-(n+1)a)^2\right]$$

Let us make the substitution $y = \sqrt{2}(x - (n+1)a) \implies \mathrm{d}y = \sqrt{2}\mathrm{d}x$ and putting it in the integral we have the following expression,

$$\frac{e^{-a^2/4\sigma^2}}{\sqrt{2\pi\sigma^2}}\int_{-\infty}^{\infty} \exp\left[-\frac{y^2}{2\sigma^2}\right] = \exp\left[-\frac{a^2}{4\sigma^2}\right]$$

Then the tunneling amplitude is estimated as:

$$\boxed{t \approx -\frac{\hbar\omega}{2}\exp\left[-\frac{a^2}{4\sigma^2}\right] = -\frac{\pi}{a}\sqrt{\frac{V_0\hbar^2}{2m}}\exp\left[-\frac{\pi}{4}\sqrt{\frac{2V_0ma^2}{\hbar^2}}\right]}$$

Note that this amplitude is exponentially falling with V_0 which is nice, since for deep enough wells, the tight-binding approximation is expected to be valid.